

JOB PORTAL MANAGEMENT SYSTEM

School of Computing
Bachelor of Technology
in
Computer Science & Engineering By

Kalaiyarasan K (VTU26102)

10211CS224
Full Stack Application Development
Slot : E3



**DEPARTMENT OF COMPUTER SCIENCE AND
ENGINEERING
SCHOOL OF COMPUTING**

**VEL TECH RANGARAJAN Dr.SAGUNTHALA R&D
INSTITUTE OF SCIENCE AND TECHNOLOGY
(Deemed to be University Estd u/s 3 of UGC Act, 1956)
CHENNAI 600 062, TAMILNADU, INDIA**

May, 2026

BONAFIDE CERTIFICATE

It is certified that the work contained in the Full Stack Application Development report titled "Job Portal Management System" by Kalaiyarasan K (VTU26102) has been carried out in Winter semester of Academic year 2025-2026(WS2526).

Signature of Faculty

Dr.HAFIZUR RAHMAN.

Assistant Professor (Senior Grade)

Computer Science & Engineering

School of Computing Vel

Tech Rangarajan Dr.Sagunthala R&D Institute

of Science and Technology

May, 2026

Signature of Course Co-ordinator

Dr. Sumithra M

Assistant Professor(Senior Grade)

Computer Science & Engineering

School of Computing

Vel Tech Rangarajan Dr.Sagunthala R&D

Institute of Science and Technology

May, 2026

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ABSTRACT

The traditional recruitment process often involves manual methods, leading to inefficiencies in job searching, profile management, and application tracking. This project presents Job Portal Management System, a web-based application designed to streamline and automate the recruitment process for both job seekers (students) and employers.

The system is developed using Spring Boot + Spring MVC with Thymeleaf frontend, enabling secure authentication, efficient database management via Spring Data JPA, and smooth data handling through REST APIs. Employers can post and manage job listings with details such as title, description, skills required, and salary, while job seekers can search/filter by category/location/experience, apply online, upload resumes, and track application status through an intuitive interface.

The application ensures data integrity and prevents duplicate applications using @NotNull/@Size validation. It also implements Spring Security role-based access control (Student/Employer/Admin) with password hashing for enhanced security. Overall, the system improves recruitment efficiency, enhances user experience, and provides a reliable platform for job management.

Keywords: Job Portal Management System, Spring Boot, Spring Security, Recruitment System, Role-Based Access, Web Application.

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LIST OF ACRONYMS AND ABBREVIATIONS

API	Application Programming Interface
CRUD	Create, Read, Update, Delete
DB	Database
MVC	Model View Controller
JPA	Java Persistence API
ORM	Object Relational Mapping
SQL	Structured Query Language
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
OTP	One-Time Password
HTTP	HyperText Transfer Protocol
UI	User Interface
UX	User Experience
Auth	Authentication
JWT	JSON Web Token

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Chapter 1

INTRODUCTION

1. Introduction

In the current era of digital transformation, recruitment processes are rapidly evolving with technology-driven solutions. Traditional hiring methods rely on manual processes, offline applications, and fragmented communication, leading to inefficiencies and delays. A centralized automated system connecting job seekers (students) and employers has become essential.

The Job Portal Management System is a web-based platform that streamlines recruitment by providing a unified interface for students and employers. It enables profile creation, resume uploads, job search by category/location/ experience, and online applications, while employers manage job postings and candidate applications efficiently.

1. Problems in Existing System

Traditional systems lack centralization, have inefficient communication, and struggle managing large application volumes. Job seekers can't easily find relevant opportunities while employers face difficulties filtering/shortlisting candidates. Manual verification creates security risks and fake registrations. The absence of real-time updates and tracking makes recruitment inefficient.

2. Proposed System

The Job Portal Management System overcomes existing limitations with a centralized Spring Boot platform. Students register, maintain profiles, upload resumes via MultipartFile, search/filter jobs, and track applications. Employers

post jobs (title, description, skills, salary), review applications, and shortlist/reject candidates through dedicated dashboards and REST APIs.

3. Technology Used

Built with Spring Boot + Spring MVC for REST APIs using `@RestController/ @Controller`, Thymeleaf/HTML5/CSS3/JavaScript for responsive frontend. Spring Data JPA manages User/Job/Application entities with relationships, MySQL/H2 database, Spring Security for role-based access (Student/Employer/Admin) with password hashing.

4. System Features

Role-based access control differentiates students, employers, administrators. Resume upload via `MultipartFile`, job search/filtering by category/ location/-experience, real-time application tracking, dashboard analytics for applications/shortlists. Custom JPA queries, `@NotNull/@Size` validation, responsive forms with JavaScript dynamic tables enhance usability.

5. System Benefits

Automates recruitment, reduces manual work, improves efficiency via REST APIs and JPA relationships. Employers manage postings/applicants easily while students access relevant jobs with status tracking. Spring Security, data validation, and JPA relationships ensure data integrity for production-ready recruitment applications.

1.2 Aim of the Project

The primary aim of the Job Portal Management System is to develop an efficient and secure platform that simplifies the recruitment process in a digital environment. The system is designed to enhance interaction between job seek-

ers (students) and employers while ensuring accuracy, transparency, and ease of use through Spring Boot REST APIs.

1. Objectives

The project focuses on providing a centralized Spring Boot solution integrating all recruitment aspects into a single platform. The specific objectives are:

- Design/develop centralized job portal for efficient recruitment management
- Enable students to create profiles, upload resumes via `MultipartFile`, apply easily
- Allow employers to post jobs (title/description/skills/salary) and manage applications
- Implement Spring Security role-based access (Student/Employer/Admin)
- Provide job search/filtering by category/location/experience via JPA queries
- Enable real-time application status tracking with notifications
- Incorporate role-based dashboards and custom JPA finder methods
- Ensure data integrity through `@NotNull/@Size` validation and JPA constraints

By achieving these objectives, the system improves recruitment efficiency and provides seamless experience for students and employers through responsive Thymeleaf interface and professional Spring Boot architecture.

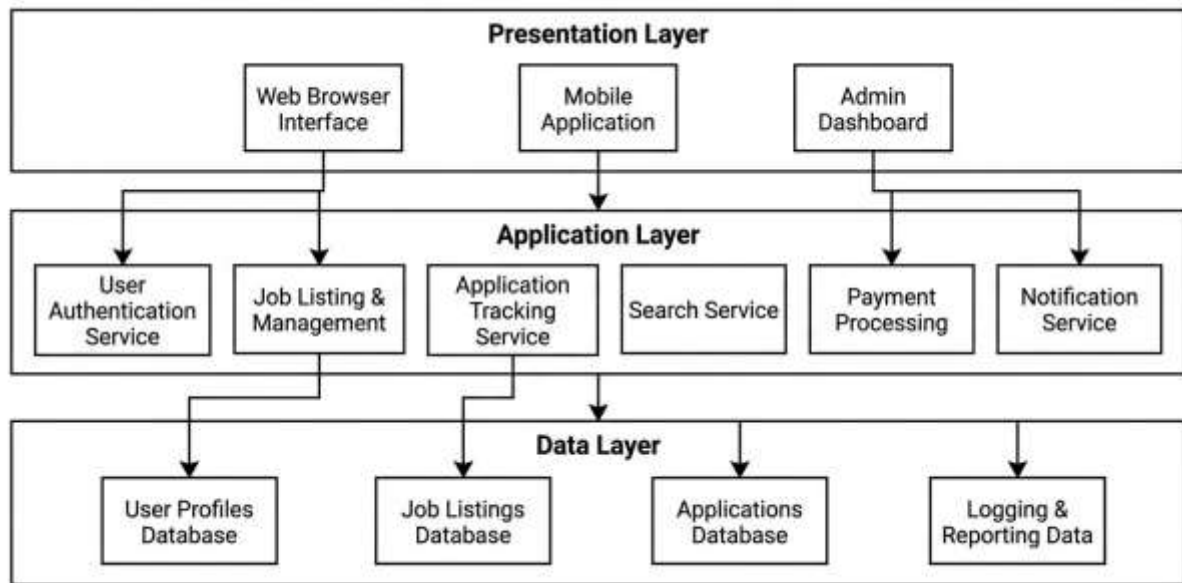


Figure 1.1: Architecture of the Job Portal Management System

Figure 1.1 illustrates the layered system architecture of the Job Portal Management System, showing the interaction between the presentation, application, and data layers. The presentation layer provides a responsive user interface for job browsing, application submission, and dashboard access. The application layer handles core services such as authentication, job management, application tracking, search functionality, and notifications, ensuring proper execution of business logic and system security. The data layer stores and manages persistent information including user profiles, job listings, and application records. This modular architecture enhances scalability, maintainability, and efficient communication between system components, making it suitable for real-world implementation.

Chapter 2

SURVEY OF SIMILAR APPLICATIONS IN MARKET

The market for job portal systems includes various platforms ranging from basic job listing websites to advanced recruitment management systems. These platforms aim to simplify job searching, candidate management, and hiring processes for campus placements. Analysis of these applications reveals their strengths and limitations, highlighting the need for an efficient Spring Boot-based solution.

1. Generic Job Portal Platforms

Popular platforms like Naukri.com and Monster provide digital environments for job seekers and employers to connect for campus recruitment.

- **Strengths:** Support job posting, candidate search, resume uploads, and centralized application tracking with scalable infrastructure.
- **Weaknesses:** Overloaded with enterprise features unsuitable for academic projects. Lack customization for campus-specific requirements and complex role-based access for students/employers.

2. Online Recruitment Systems

Web-based recruitment systems enable employers to post jobs and students to apply through online interfaces with basic profile management.

- **Strengths:** Reduce paperwork through digital applications, maintain candidate records, and provide basic search functionality.

- **Weaknesses:** Limited advanced filtering by category/location/ experience, no real-time application status tracking, and inadequate role-based security for multi-user environments (Student/Employer/Admin).

3. Manual and Traditional Campus Hiring Methods

Many colleges still use manual processes like physical resume collection, notice boards, and Excel-based shortlisting for campus placements.

- **Strengths:** Simple implementation with no technical dependencies and familiar to placement officers.
- **Weaknesses:** Time-consuming manual data entry, no real-time application tracking, error-prone shortlisting, and inefficient for managing large student applicant volumes during campus drives.

4. Gap Analysis and Proposed System Advantage

A comprehensive comparative analysis of commercial recruitment solutions reveals fundamental architectural deficiencies across multiple critical dimensions. Most platforms fundamentally lack fully integrated job posting capabilities with complete CRUD operations, real-time application lifecycle tracking spanning submission through shortlisting/rejection workflows, granular role-based security implementation featuring distinct Student/Employer/Admin access differentiation, sophisticated multi-parameter campus-specific filtering mechanisms supporting category/location/experience-based queries, production-grade input validation frameworks with @NotNull/@Size annotations, and cohesive JPA entity relationship management (One-to-Many Employer→Jobs, Many-to-One Application→Job/User) within a single, meticulously architected Spring Boot platform specifically engineered for academic recruitment automation,

scalable student placement management requirements, and institutional deployment standards.

Feature	Manual	Recruitment Tools	Job Portals	Proposed System
Job Posting	Limited	Yes	Yes	Yes
Online Application	No	Yes	Yes	Yes
Duplicate Prevention	No	No	Partial	Yes
Centralized Data	No	Partial	Yes	Yes
Role-based Access	No	Limited	Partial	Yes
JPA Filtering	No	No	Limited	Yes
Resume Upload	No	Partial	Yes	Yes
Application Analytics	No	Limited	Yes	Yes
Spring Security	No	Partial	Partial	Yes

Table 2.1: Comparative Analysis of Recruitment Systems

Table 2.1 compares existing systems against the proposed Job Portal Management System. Manual systems lack digital capabilities while commercial portals offer partial solutions unsuitable for academic environments. The Spring Boot system provides comprehensive integration of REST APIs, JPA relationships, Thymeleaf dashboards, and Spring Security, ensuring production-ready recruitment automation with campus-specific customization, advanced filtering, and complete application lifecycle management.

Chapter 3

FRAMEWORK DESCRIPTION

1. Frontend Implementation

The frontend of the Job Portal Management System uses modern web technologies to provide responsive interfaces for students and employers through Thymeleaf.

1. Environment Setup

- **Frontend:** HTML, CSS, JavaScript, Thymeleaf
- **IDE:** IntelliJ IDEA / VS Code
- **Browser:** Chrome or Edge

2. Project Structure

- `/templates`: Thymeleaf HTML pages
- `/static`: CSS, JavaScript, images
- `/controllers`: Request mapping
- `/services`: Business logic
- `/repository`: Database layer

3. Execution Procedure

1. Configure project in IDE
2. Run Spring Boot application
3. Access: <http://localhost:8080>

```

Spring Boot (v4.8.5)

326-85-05T17:37:26.258+05:30 [INFO] --- [job-portal] ---
326-85-05T17:37:26.259+05:30 [INFO] --- [job-portal] ---
326-85-05T17:37:26.382+05:30 [INFO] --- [job-portal] ---
326-85-05T17:37:28.479+05:30 [INFO] --- [job-portal] ---
326-85-05T17:37:29.694+05:30 [INFO] --- [job-portal] ---
326-85-05T17:37:29.732+05:30 [INFO] --- [job-portal] ---
326-85-05T17:37:29.733+05:30 [INFO] --- [job-portal] ---
326-85-05T17:37:29.804+05:30 [INFO] --- [job-portal] ---
326-85-05T17:37:30.310+05:30 [INFO] --- [job-portal] ---
326-85-05T17:37:30.321+05:30 [INFO] --- [job-portal] ---
326-85-05T17:37:31.830+05:30 [INFO] --- [job-portal] ---
326-85-05T17:37:32.006+05:30 [INFO] --- [job-portal] ---
326-85-05T17:37:32.906+05:30 [INFO] --- [job-portal] ---
326-85-05T17:37:32.994+05:30 [INFO] --- [job-portal] ---
326-85-05T17:37:33.836+05:30 [INFO] --- [job-portal] ---

Database driver: H2 JDBC Driver
Database dialect: H2Dialect
Database version: 2.4.204
Default catalog/schema: JOBPORTALDB/PUBLIC
Autocommit mode: undefined/unknown
Isolation level: READ_COMMITTED [default READ_COMMITTED]
JDBC fetch size: 100
Pool: DataSourceConnectionProvider
Minimum pool size: undefined/unknown

main] com.jobportal.JobPortalApplication : Starting JobPortalApplication using Java 21.0.9 wit
main] com.jobportal.JobPortalApplication : No active profile set, falling back to 1 default pr
main] org.springframework.data.jpa.repository.config.JpaRepositoryConfigDelegate : Bootstrapping Spring Data JPA repositories in DEFAL
main] org.springframework.data.jpa.repository.config.JpaRepositoryConfigDelegate : Finished Spring Data repository scanning in 143 ms.
main] org.springframework.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
main] org.springframework.boot.tomcat.TomcatWebServer : Starting service [Tomcat]
main] org.springframework.boot.tomcat.TomcatWebServer : Starting Servlet engine: [Apache Tomcat/11.0.10]
main] org.springframework.boot.tomcat.TomcatWebServer : Root WebApplicationContext: initialization complete
main] org.hibernate.orm.jpa : Processing PersistenceUnitInfo [name: de
main] org.hibernate.orm.jpa : HH000000: Hibernate ORM core version 7.2.7.Final
main] org.hibernate.orm.jpa : No LoadTimeWeaver setup: ignoring JPA class transfe
main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Starting...
main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Added connection conn1: url=jdbc:h2:
main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Start completed.
main] org.hibernate.orm.jpa : HH00000025: H2Dialect does not need to be specific
main] org.hibernate.orm.jpa : HH00000003: Database info.

```

Figure 3.1: Spring Boot Application Execution Output in Console

Figure 3.1 Spring Boot application startup and execution output showing successful initialization of the server and database configuration.

2. Backend Implementation

The backend uses Spring Boot and Spring MVC for robust RESTful web services with complete CRUD operations across all entities.

1. Core Features

- REST API development with `@RestController/@Controller`
- Role-based authentication (Student, Employer, Admin)
- Spring Security with password hashing
- Business logic via `@Service` layer

2. Execution Procedure

- Configure application.yml properties
- Connect MySQL/H2 database

table, which stores important user details such as user ID, name, email, encrypted password, phone number, resume path, and user role. The database is structured to manage user information efficiently and support features like authentication, job applications, and role-based access. This design helps ensure data integrity, security, and fast retrieval of records.

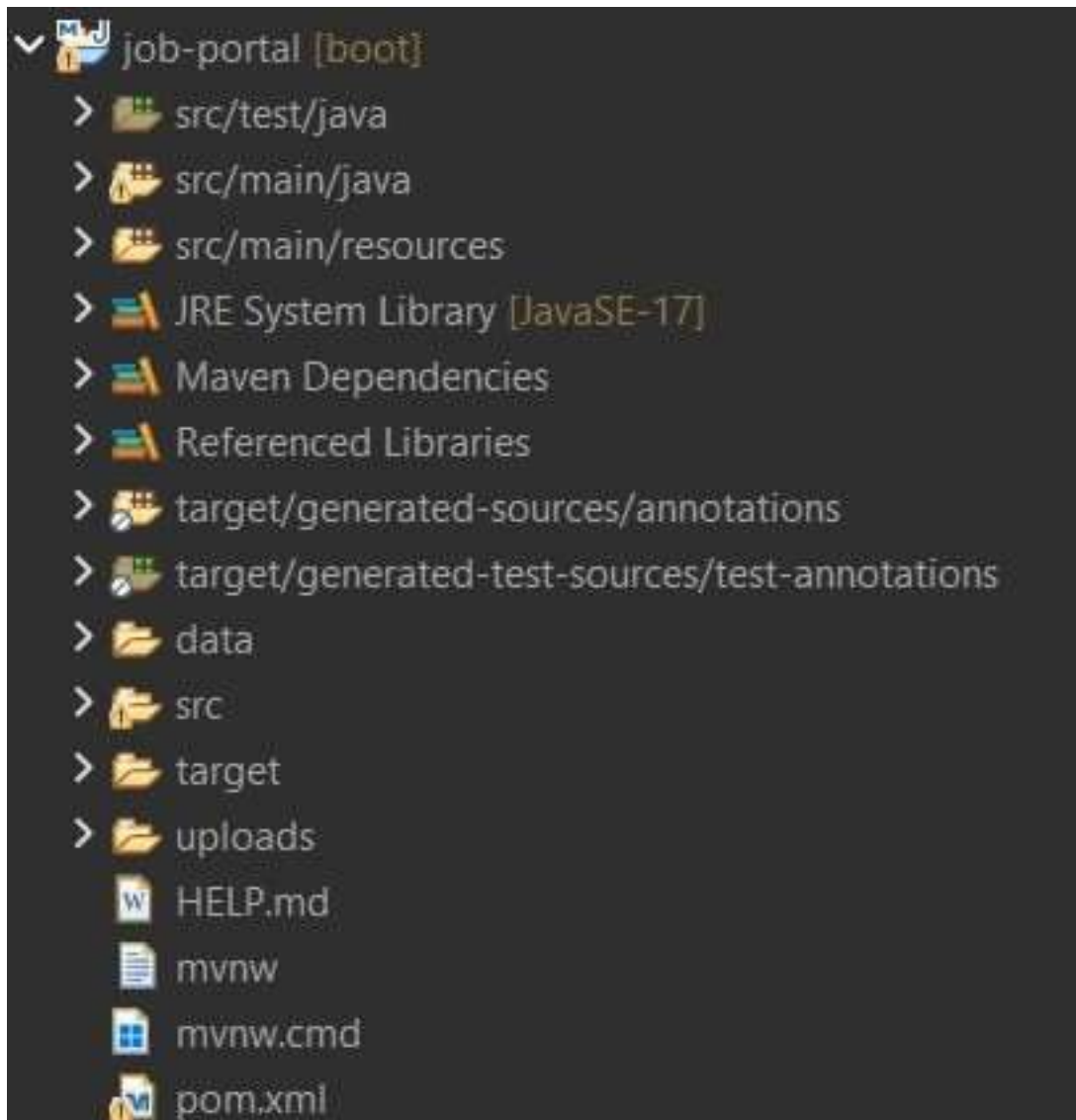


Figure 3.3: Spring boot project structure

Figure 3.3 shows the project structure of the Job Portal Management System, illustrating the organization of backend, frontend, and configuration components.

4. Security Features

The system incorporates comprehensive Spring Security mechanisms ensuring safe data handling, controlled access, and protection against common vulnerabilities across student, employer, and admin modules with enterprise-grade authentication.

- **Authentication:** Spring Security with JWT/session management and CSRF protection
- **Role-Based Access:** Granular Student/Employer/Admin permissions via `@PreAuthorize`
- **Password Encryption:** BCryptPasswordEncoder with salting and 12+ rounds
- **Input Validation:** `@NotNull/@Size/@Valid` annotations with custom validators
- **Exception Security:** `@ControllerAdvice` preventing stack trace exposure

5. System Technology Stack Overview

Component	Technology	Description
Frontend	HTML5, CSS3, Thymeleaf	Responsive role-specific UI
Backend	Spring Boot + MVC	REST APIs, business logic
Database	MySQL / H2	User/Job/Application storage
ORM	Spring Data JPA	Entity queries, relationships
Security	Spring Security	Authentication, authorization
Validation	Bean Validation	<code>@ControllerAdvice</code> handling

Table 3.1: Job Portal Management System Technology Stack

Table 3.1 presents the production-grade technology stack where Spring Boot orchestrates REST APIs through `@RestController` endpoints, Thymeleaf pro-

cesses role-specific templates, JPA manages complex One-to-Many/Many-to-One entity relationships, and Spring Security enforces comprehensive multi-user access control with industry-standard encryption protocols.

Chapter 4

METHODOLOGIES

1. Project Architecture

The architecture of the Job Portal Management System follows a multi-tier Spring Boot web design ensuring scalability, security, and maintainability across student, employer, and administrator roles, with clear separation of concerns for production readiness.

- **Presentation Layer (Frontend):** Built using HTML5, CSS3, JavaScript ES6, and Thymeleaf integrated with Spring Boot MVC. It provides responsive, role-based interfaces such as student job search dashboards, employer job management panels, and admin monitoring views. Features include dynamic filtering, form handling, real-time updates, and responsive navigation using Thymeleaf fragments and client-side scripts.
- **Application Layer (Backend Services):** Implemented using Spring Boot 3.x and Spring MVC, exposing RESTful APIs via `@RestController` and `@Controller` for CRUD operations on User, Job, and Application entities. The service layer manages workflows like application processing, role-based access using Spring Security, validation (`@Valid`, `@NotNull`), exception handling (`@ControllerAdvice`), and transaction management.
- **Data Layer (Database Persistence):** Uses MySQL for production and H2 for development to store user profiles, job postings, and application data. Spring Data JPA with Hibernate ORM handles entity relationships (One-to-Many, Many-to-One), supports search queries, and ensures data integrity.

and optimized performance.

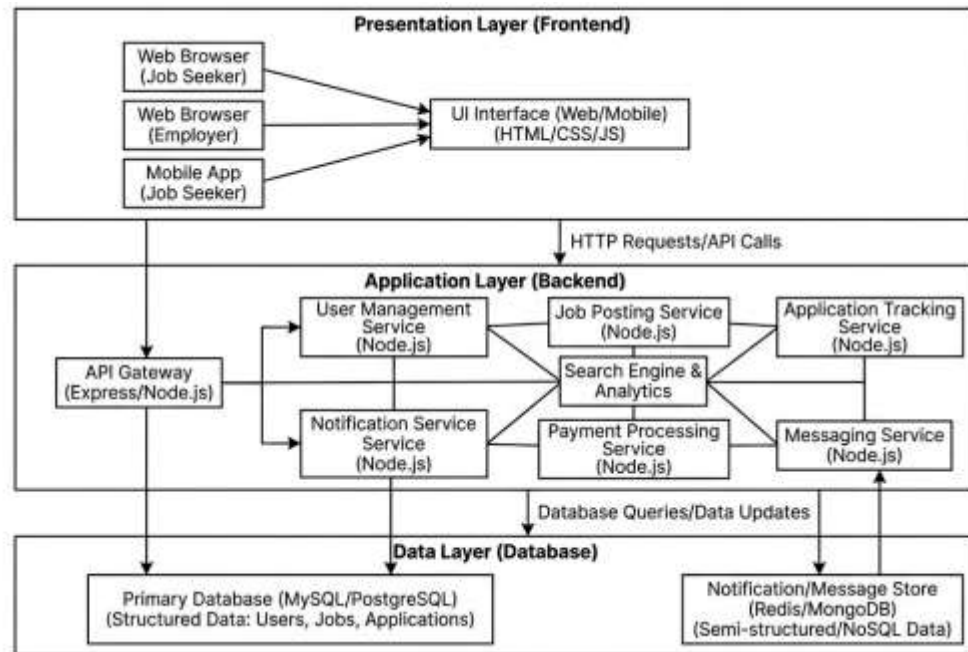


Figure 4.1: System Architecture of Job Portal Management System

Figure 4.1 shows the layered system architecture of the Job Portal Management System, illustrating the overall interaction between the presentation layer, application layer, and database layer. The presentation layer provides the user interface through web and mobile access, allowing job seekers and employers to interact with the system using HTML, CSS, JavaScript, and Thymeleaf. The application layer processes requests through backend services such as user management, job posting, application tracking, search, notifications, and payment-related operations. The database layer stores structured information including user profiles, job listings, applications, and notification data, ensuring reliable persistence and retrieval. This layered architecture improves modularity, maintainability, scalability, and communication between components, making the system efficient and easy to manage.

4.2 System Workflow and Data flow

The System Workflow Diagram provides a comprehensive visual representation of the Job Portal Management System's operational flow, illustrating seamless interaction patterns between authenticated users (students, employers, administrators) and core system components throughout the complete campus recruitment lifecycle.

The process initiates at the responsive landing page where users authenticate via Spring Security-enabled login forms requiring valid credentials, followed by role-specific session establishment and secure redirection to personalized Thymeleaf dashboards based on User entity role attributes (Student/Employer/Admin). Job seekers access advanced search functionality with multi-parameter filtering by category, location, and experience requirements, upload resumes through secure MultipartFile endpoints, and submit applications creating Application entity records linked via JPA Many-to-One relationships to respective Job and User entities.

Employers utilize dedicated management panels to perform complete CRUD operations on Job entities (title, description, required skills, salary range), review applicant lists with real-time status tracking, execute shortlisting/rejection workflows updating Application status fields atomically through Spring transaction management, and access dashboard analytics displaying application volume, shortlist ratios, and placement metrics. All backend operations leverage RESTful API endpoints communicating with Spring Data JPA repositories ensuring data persistence in MySQL/H2 databases while @ControllerAdvice handles exceptions globally and custom finder methods optimize query performance for production-scale campus recruitment operations.

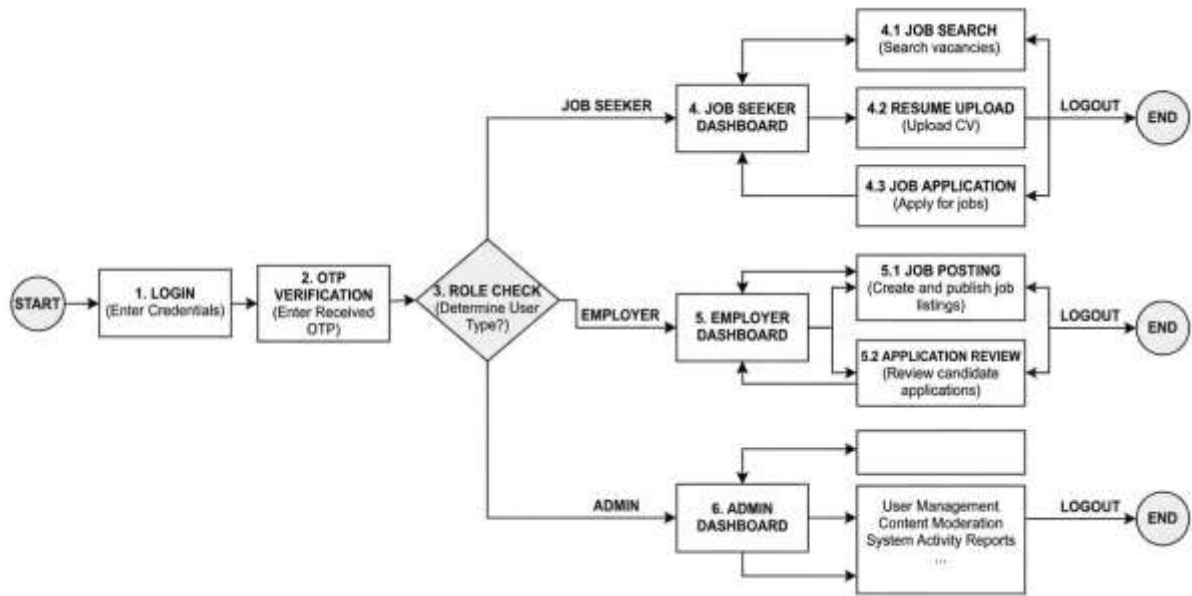


Figure 4.2: System Workflow Diagram

Figure 4.2 shows the system workflow of the Job Portal Management System. The process begins when the user accesses the landing page and enters login credentials. The system verifies the credentials and sends an OTP for secure authentication, which the user must validate before proceeding. After successful verification, the system checks the user role and redirects the user to the appropriate dashboard. If the user is a job seeker, they are directed to the student dashboard where they can search for jobs, upload resumes, and apply for vacancies. If the user is an employer, they are redirected to the employer dashboard where they can create job postings, edit or delete listings, review applications, and shortlist candidates. If the user is an admin, they are taken to the admin panel for managing users, jobs, and system activity. This workflow ensures a secure, organized, and efficient recruitment process within the system.

4.2.1 DFD Level 0: Context Diagram

The Level 0 diagram represents the Job Portal Management System as a single centralized process interacting with external entities including Student, Em-

ployer, and Administrator through defined data flows for complete recruitment automation.

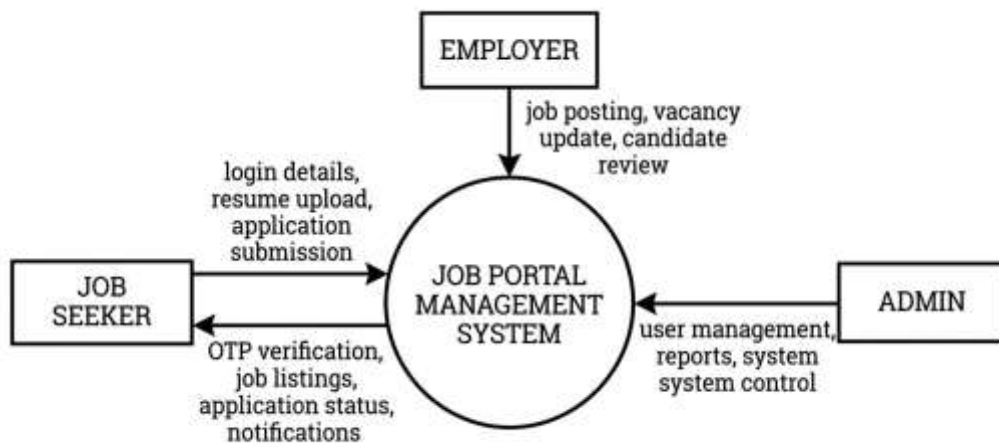


Figure 4.3: DFD Level 0: Context Diagram

Figure 4.3 shows the Level 0 Data Flow Diagram of the Job Portal Management System. It represents the overall interaction between external entities such as Job Seeker, Employer, and Admin with the system. The diagram highlights major data exchanges including job listings, application submissions, resume uploads, and OTP-based authentication. This diagram provides a high-level overview of how data flows within the system and how the main users communicate with the platform.

4.2.2 DFD Level 1: Functional Decomposition

The Level 1 diagram breaks down the system into internal processes such as Spring Security authentication, job posting/editing, application submission with resume upload, status tracking, and employer shortlisting workflows.

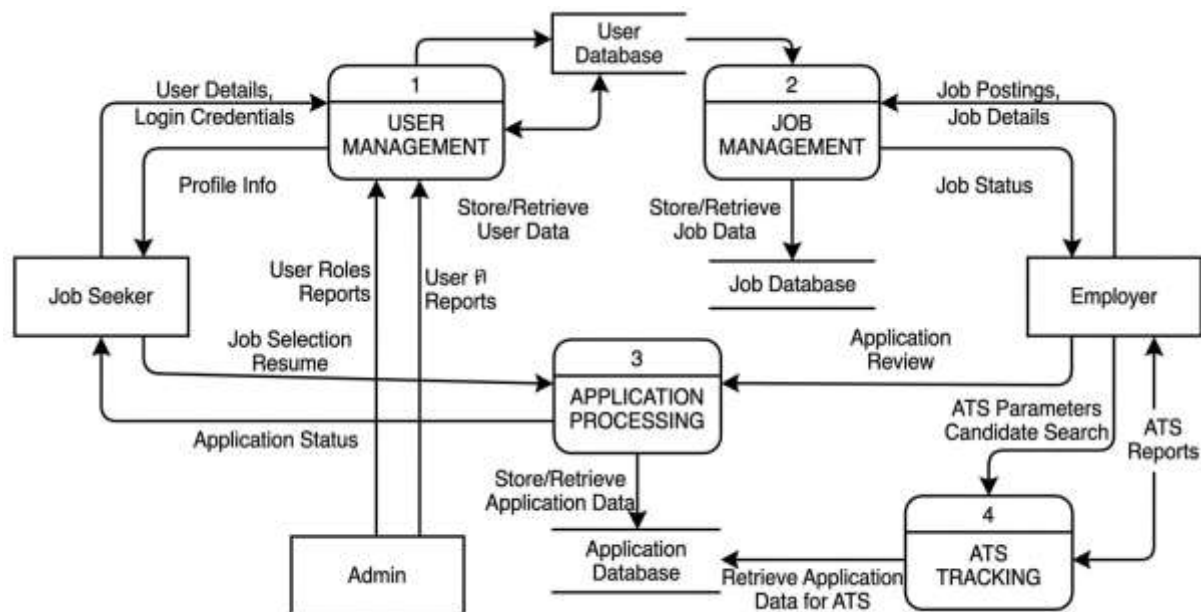


Figure 4.4: DFD Level 1

Figure 4.4 shows the Level 1 Data Flow Diagram of the Job Portal Management System. It gives a detailed view of the internal processes, including User Management, Job Management, Application Processing, and ATS Tracking. The diagram also shows how data is stored in and retrieved from the database. This level helps explain the system's internal working and data handling in a clearer way.

3. System Modules and Functional Architecture

The system is organized into modular components ensuring maintainability and efficient Spring Boot execution across all user roles.

1. Module 1: User Authentication and Role Management

Handles Spring Security login and role-based access control.

Functionality:

- Student/Employer/Admin registration and authentication
- Password hashing with BCrypt encoder

- Role-based dashboard redirection

2. **Module 2: Job Posting and Management**

Employer module for complete job CRUD operations.

Functionality:

- Create/update/delete jobs (title, skills, salary, category)
- JPA One-to-Many relationship with Employer entity
- Job visibility and status management

3. **Module 3: Job Search and Filtering**

Advanced search capabilities for students.

Functionality:

- Multi-parameter filtering (category/location/experience)
- Custom JPA finder methods and indexed queries
- Dynamic table sorting with JavaScript

4. **Module 4: Application Tracking System**

Manages complete application lifecycle.

Functionality:

- Job application submission with resume upload
- Real-time status tracking (Pending/Shortlisted/Rejected)
- Duplicate prevention via unique constraints

5. **Module 5: Resume Upload and Profile**

Student profile and document management.

Functionality:

- MultipartFile resume upload to database/local storage
- Profile CRUD operations
- Resume download for employer review

6. Module 6: Employer Dashboard and Analytics

Employer analytics and applicant management.

Functionality:

- Applicant list with filtering and search
- Shortlist/reject workflows with status updates
- Application count and shortlist ratio metrics

7. Limitations

- Requires stable internet connectivity
- Basic analytics (no advanced ML features)
- Performance tied to server resources

Chapter 5

RESULTS AND DISCUSSIONS

1. System Implementation Overview

The Job Portal Management System is designed as a comprehensive web-based platform integrating multiple modules for both job seekers and employers.

- Landing Page Interface
- User Authentication with OTP
- Employer Dashboard
- Job Posting Module
- Job Listings Interface
- Application Tracking System
- Resume Upload Module
- Notification System
- Analytics Module
- Database Layer

The system follows a modular architecture ensuring scalability, efficient communication, and smooth management of recruitment-related activities.

5.2 System Interface Visualizations

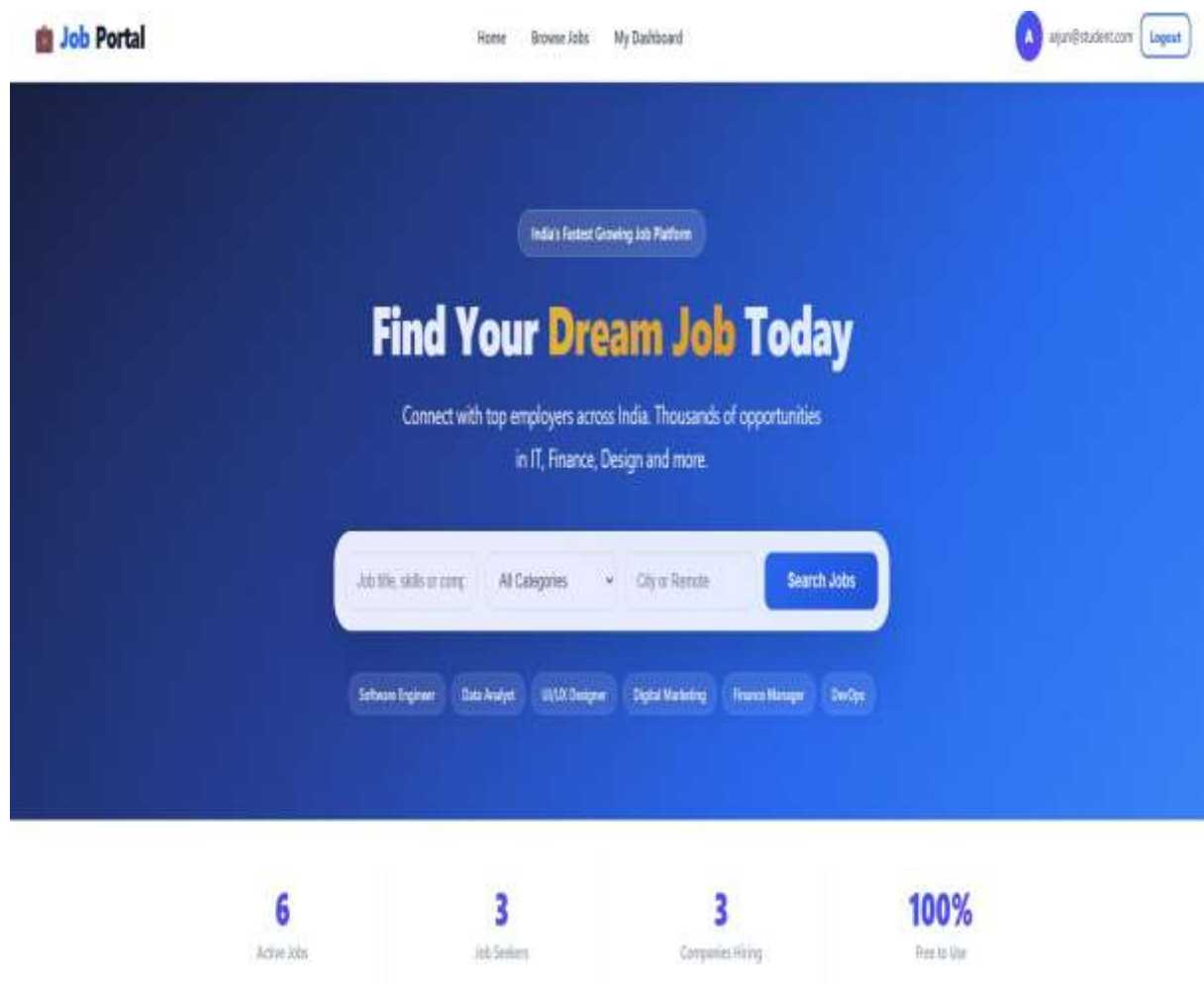


Figure 5.1: Landing Page

Figure 5.1 shows the Landing Page of the Job Portal system, which acts as the main entry point for users after login. It allows users to search for jobs by title, category, and location, and highlights key statistics such as active jobs, registered job seekers, and companies hiring. The layout clearly showcases the core purpose of the portal—helping users quickly find suitable opportunities—while the top navigation provides direct access to job browsing and the user dashboard. Overall, the design focuses on clarity, prominent search functionality, and an intuitive structure that supports efficient job discovery.

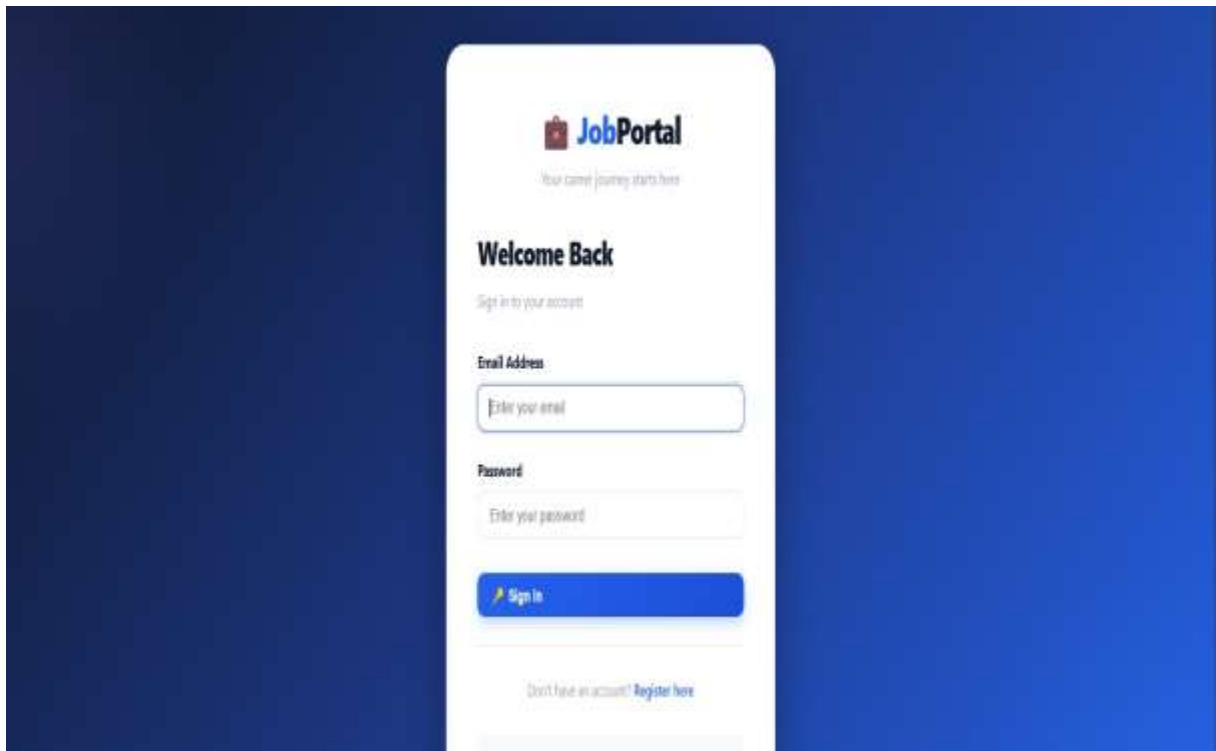


Figure 5.2: User Login and OTP Verification

Figure 5.2 shows the Login Page of the Job Portal system, which serves as the primary access point for job seekers, employers, and administrators to enter the application. It provides clearly labeled input fields for email address and password, along with a prominent “Sign In” button that facilitates a straightforward and secure login process. The interface is kept clean and user-friendly, reducing confusion and helping users quickly understand how to proceed. Validation messages and prompts can be displayed near the fields to inform users about missing or incorrect data, thereby improving accuracy during authentication. Overall, the login page balances usability and security, ensuring that only authorized users can continue to their respective dashboards.

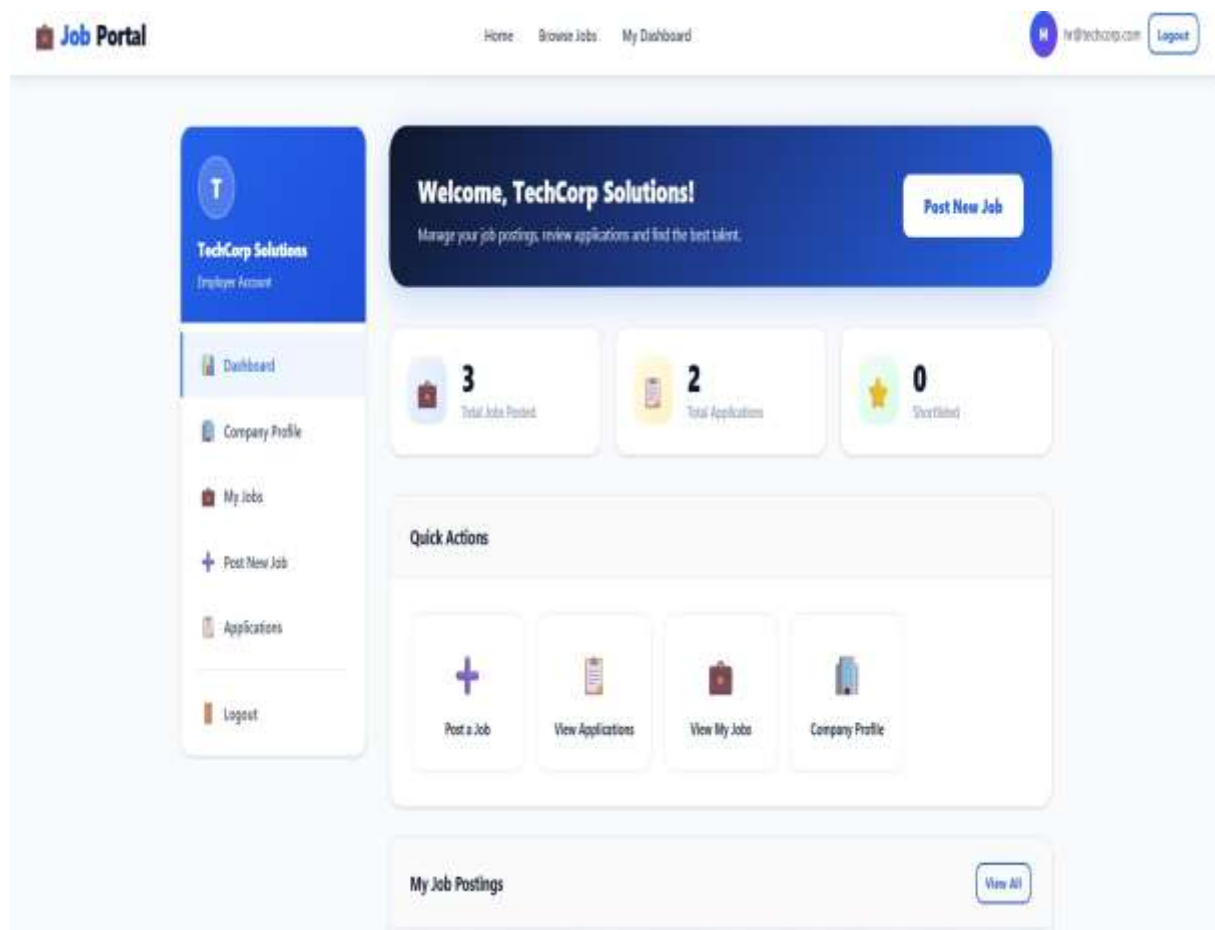


Figure 5.3: Employer Dashboard

Figure 5.3 shows the Employer Dashboard of the Job Portal system, which acts as the central workspace for company users to manage all their recruitment activities after logging in. It displays a personalized welcome panel along with key summary metrics such as the total number of jobs posted, total applications received, and shortlisted candidates, giving employers an instant overview of their hiring status. The left-side navigation menu provides structured access to core features like the company profile, list of jobs, posting new jobs, and viewing applications, ensuring that important options are always one click away. Quick-action tiles in the main area further simplify frequent tasks such as creating a new job or reviewing applications, thereby reducing the time required to perform routine operations.

The screenshot displays the 'Job Posting Interface' within a 'Job Portal' system. The interface is organized into a grid of input fields for creating a job listing. At the top, there is a navigation bar with links for 'Home', 'Browse Jobs', and 'My Dashboard', along with a user profile icon and email address 'hr@techcorp.com'. The main form area contains the following fields:

- Job Title ***: A text input field with a placeholder 'e.g. Java Backend Developer'.
- Company Name ***: A text input field with a placeholder 'Your company name'.
- Job Description ***: A large text area with a placeholder 'Describe the role, responsibilities, and what you're looking for...' and a character count '1 / 1000 characters'.
- Category ***: A dropdown menu with 'Select category'.
- Job Type**: A dropdown menu with 'Select type'.
- Location ***: A text input field with a placeholder 'e.g. Bangalore, Remote'.
- Salary Range**: A text input field with a placeholder 'e.g. 4 LPA - 8 LPA'.
- Experience Required**: A dropdown menu with 'Any experience'.
- Education Required**: A text input field with a placeholder 'e.g. B.Tech / MCA'.
- Number of Vacancies**: A text input field with a placeholder 'e.g. 0'.
- Last Date to Apply**: A date picker field with a placeholder 'dd-mm-yyyy'.
- Skills Required**: A text input field with a placeholder 'e.g. Java, Spring Boot, MySQL, Git' and a note 'Separate skills with commas'.

At the bottom of the form, there are two buttons: 'Post Job' (highlighted in blue) and 'Cancel'.

Figure 5.4: Job Posting Interface

Figure 5.4 shows the Job Posting Form used by employers in the Job Portal system to create and manage detailed job listings. The form includes structured input fields for key information such as job title, company name, job description, category, job type, location, salary range, required experience, education, number of vacancies, last date to apply, and skills required. Its organized layout ensures that all essential data is captured accurately while keeping the interface easy to understand and fill out. Overall, this form streamlines the job posting process, enforces validation on mandatory fields, and enables employers to publish clear and standardized job opportunities on the platform.

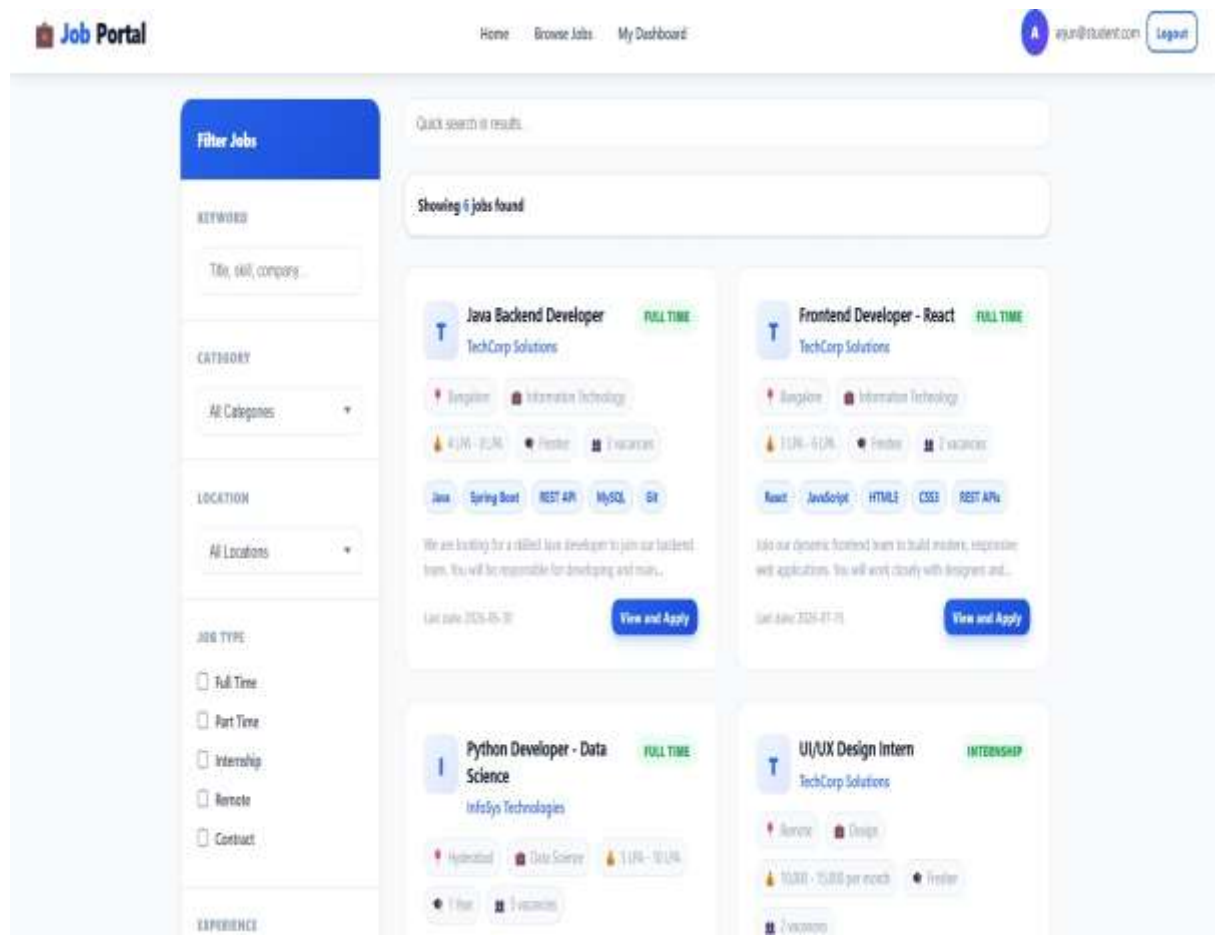


Figure 5.5: Job Details and Application Page

Figure 5.5 shows the Browse Jobs page of the Job Portal system, where job seekers can view multiple job listings along with relevant details and application options. The interface displays opportunities in a structured card-based layout, with each job highlighting key information such as title, company name, location, salary range, experience level, and required skills. A filter panel on the left allows users to refine results by keyword, category, location, job type, and experience, making it easier to focus on matching roles. Interactive elements like the “View and Apply” button on each card enable users to quickly open a job description and submit their application. Overall, the page offers a clean, responsive, and user-friendly environment that supports efficient job discovery and comparison.

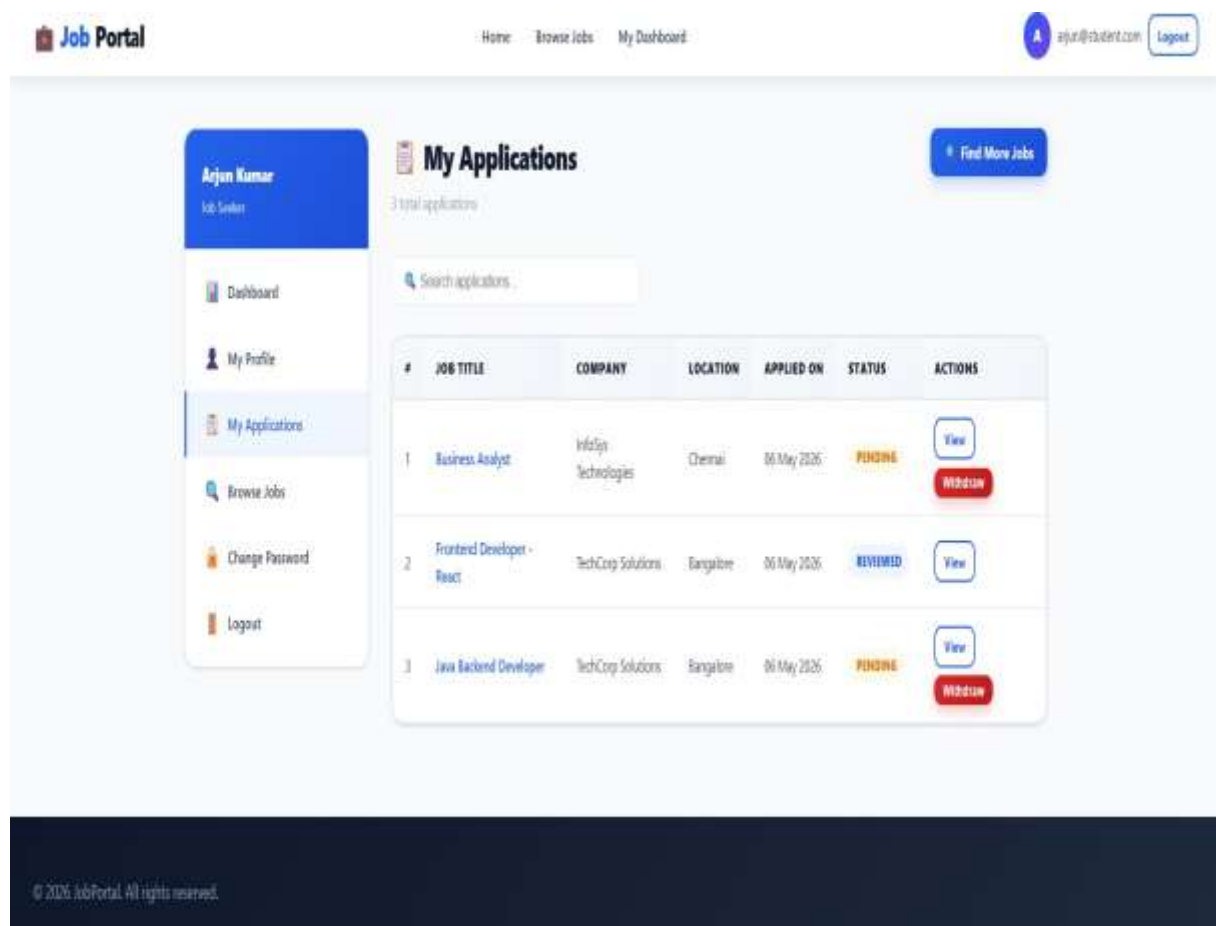


Figure 5.6: Application Tracking System

Figure 5.6 shows the “My Applications” page of the Job Portal system, which functions as an output and tracking interface for job seekers. It lists all submitted applications in a tabular format with columns for job title, company, location, applied date, and current status, giving users a clear overview of their progress for each role. A search bar allows quick filtering of applications, while action buttons such as “View” and “Withdraw” enable users to inspect details or cancel applications when needed. The left-side navigation menu provides access to other key sections like the dashboard, profile, job browsing, and password management, ensuring smooth movement across the portal. Overall, this page helps candidates monitor their job applications in an organized way and supports better decision-making during the job search process.

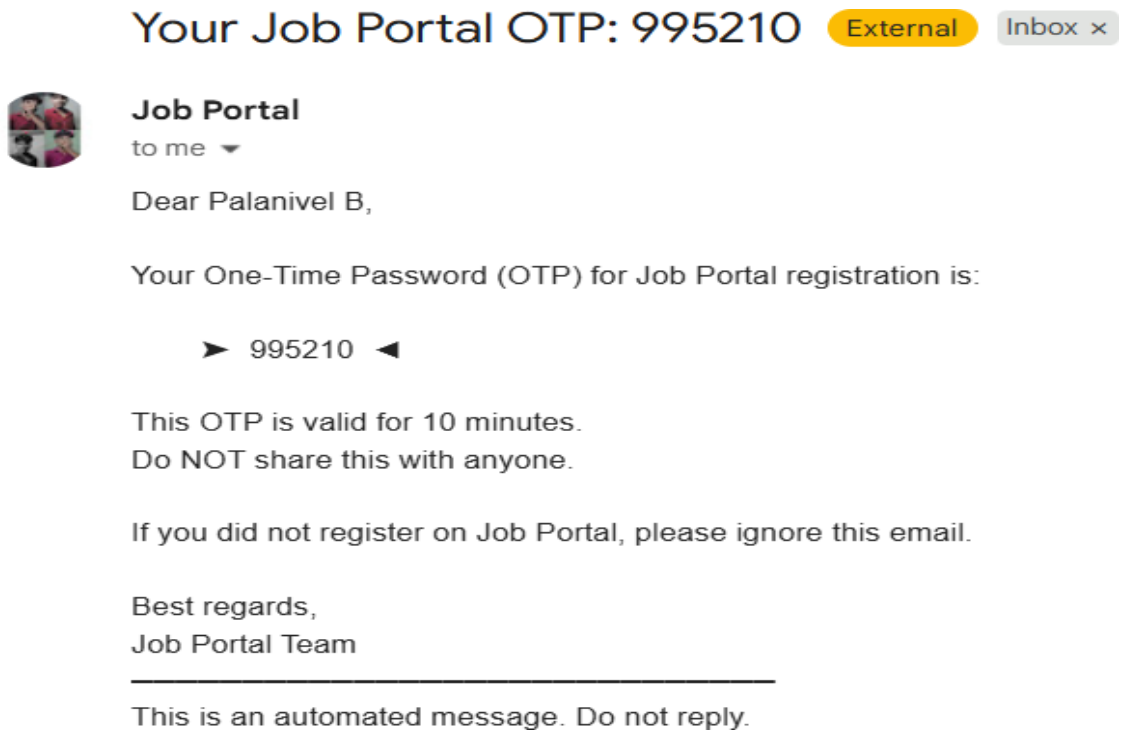


Figure 5.7: OTP Verification Process

Figure 5.7 shows the OTP verification email generated by the Job Portal system and sent to the user during registration. The system creates a unique One-Time Password (OTP) and delivers it to the registered email address to confirm the user's identity before account activation. The message clearly displays the OTP value, its validity period, and security instructions such as not sharing the code with others, thereby reducing the risk of unauthorized access. If the recipient did not initiate registration, the email instructs them to ignore the message, adding an extra layer of safety against misuse. Overall, this OTP mechanism strengthens authentication and ensures that only legitimate users can complete the registration process.

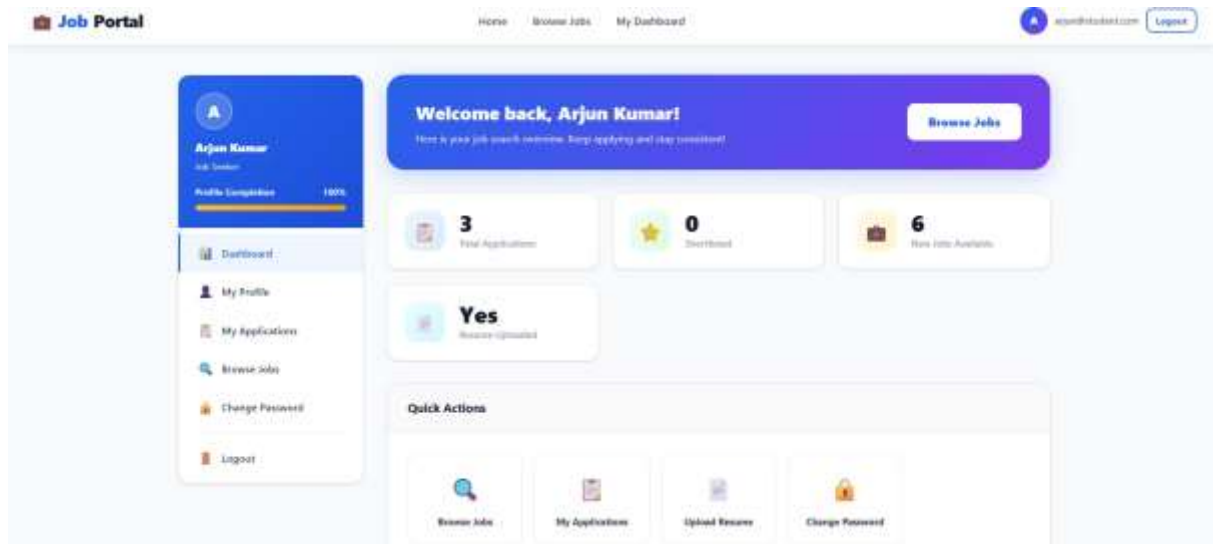


Figure 5.8: Job Seeker Dashboard

Figure 5.8 shows the Job Seeker Dashboard of the Job Portal system, which provides a centralized interface for candidates to manage their job search activities after logging in. The left-side panel displays the user’s profile card with name, role, and profile completion status, along with navigation links to sections such as My Profile, My Applications, Browse Jobs, Change Password, and Logout. The main dashboard area highlights key summary metrics including total applications submitted, number of shortlisted jobs, new jobs available, and whether a resume has been uploaded, giving users a quick overview of their current status. A prominent welcome banner and “Browse Jobs” button encourage users to continue exploring opportunities. Quick-action tiles at the bottom further simplify frequent tasks like viewing applications or uploading a resume, making this dashboard an efficient starting point for managing all career-related activities within the portal.

3. Module Explanations

1. Landing Page / Job Listings

The landing page acts as the entry point of the system, displaying available job opportunities with key details such as title, company, and location. It enables users to browse, search, and identify suitable jobs based on their preferences.

2. User Authentication with OTP

The authentication module provides secure access using OTP-based verification. It ensures only authorized users can log in and supports role-based access control.

3. Employer Dashboard

The employer dashboard offers a centralized interface for managing job postings and viewing applicant data, allowing efficient tracking of recruitment activities.

4. Job Posting Module

This module allows employers to create and manage job listings by entering details such as job title, description, required skills, and salary, ensuring structured job management.

5. Job Details and Application

The job details page provides complete information about a job and enables users to apply directly with proper validation, preventing duplicate applications.

6. Application Tracking System

This system allows users to monitor application status and enables employers to review and shortlist candidates, improving transparency.

7. OTP Verification System

The OTP verification mechanism ensures secure authentication during registration and login, preventing unauthorized access.

8. Database Implementation

The backend uses MySQL/H2 databases to store user data, job postings, and applications, ensuring efficient data handling and integrity through JPA.

4. Discussion

1. Automation of Recruitment Process

The system automates the recruitment lifecycle from job posting to application tracking, reducing manual effort and increasing efficiency.

2. Secure Authentication with OTP

OTP-based authentication enhances security by preventing unauthorized access and ensuring valid user verification.

3. Efficient Job Matching and Filtering

Users can quickly find relevant jobs using filtering options based on skills, location, and preferences.

4. Real-Time Application Tracking

The tracking feature allows users and employers to monitor application status in real time, improving communication.

5. User Experience Enhancement

A user-friendly interface ensures smooth navigation and better usability for both job seekers and employers.

5.4.6 Security and Data Integrity

The system ensures data protection and integrity through secure authentication, validation, and structured database management.

Chapter 6

CONCLUSION AND FUTURE ENHANCEMENTS

1. Conclusion

The Job Portal Management System successfully fulfills its primary objective of simplifying and digitizing the recruitment process through the integration of a powerful Spring Boot backend and a responsive, user-friendly web interface. The system establishes an efficient bridge between job seekers and employers, enabling smooth communication, structured data handling, and streamlined workflows. By replacing traditional manual processes with an automated digital platform, the system significantly enhances the speed, accuracy, and transparency of recruitment activities.

The project clearly demonstrates several key outcomes:

- 1. Efficient Recruitment Management:** The automation of core processes such as job posting, resume submission, application tracking, and candidate shortlisting reduces human effort and operational delays. It also minimizes the chances of errors and ensures consistency throughout the recruitment lifecycle.
- 2. Improved User Experience:** The system provides a centralized and intuitive interface where users can easily explore job opportunities, apply for positions, and monitor application status. Employers benefit from organized dashboards that simplify job and applicant management.
- 3. Secure and Scalable System:** The implementation of OTP-based authen-

tication, role-based access control, and modern development frameworks ensures that the system is secure, reliable, and capable of handling future growth and increased user demand.

Overall, the Job Portal Management System serves as a comprehensive and dependable solution for modern recruitment challenges. It not only improves operational efficiency but also enhances user satisfaction by offering a seamless and transparent hiring experience for both job seekers and employers.

2. Future Enhancements

While the current implementation delivers essential functionalities effectively, several enhancements can be introduced to further strengthen the system and expand its capabilities:

- **Advanced Analytics:** The integration of analytical dashboards can provide valuable insights into recruitment patterns, applicant behavior, and job performance metrics. This will assist employers in making data-driven hiring decisions.
- **Real-Time Notification System:** Introducing real-time notifications for job postings, application updates, interview schedules, and employer responses will improve communication and user engagement.
- **Mobile Application Development:** Creating a dedicated mobile application will enhance accessibility, allowing users to interact with the system conveniently from smartphones and tablets.
- **AI-Based Job Recommendation:** Incorporating artificial intelligence can enable personalized job recommendations based on user profiles, skills,

preferences, and previous activities, thereby improving job matching accuracy.

- **Enhanced Security Measures:** Additional security mechanisms such as multi-factor authentication, encrypted data storage, and advanced session management can be implemented to further protect user data and prevent unauthorized access.

Chapter 7

ADDRESSING COMPLEX ENGINEERING PROBLEM AND SDG

7.1 Mapping of the Project to Complex Engineering Problem (CEP)

Level	Attribute	WKs	Justification
WP1	Depth of Knowledge	WK3, WK4, WK5, WK6	Requires strong understanding of Spring Boot, Spring MVC, Spring Security, JPA, and database design to build a multi-role job portal system.
WP2	Conflicting Requirements	WK4, WK5, WK6	Balances system security with usability, ensuring smooth access for both job seekers and employers without compromising protection.
WP3	Depth of Analysis	WK3, WK4, WK5	Involves ER modeling, REST API design, and integration of modules such as file upload and email services.
WP4	Familiarity	WK4, WK8	Introduces advanced features like job matching and application tracking beyond basic CRUD operations.
WP5	Applicable Codes	WK6	Requires custom logic for resume storage, application lifecycle handling, and dynamic job filtering mechanisms.
WP6	Stakeholder Needs	WK5, WK6, WK7	Addresses requirements of multiple stakeholders including job seekers, employers, and administrators through role-based design.
WP7	Interdependence	WK3, WK5, WK6	Integrates frontend, backend, database, and security components into a unified system.

Table 7.1: Mapping of Project to Complex Engineering Problem (CEP)

Table 7.1 presents how the Job Portal Management System aligns with Complex Engineering Problems (CEP). The project demonstrates depth of knowl-

edge by applying modern frameworks such as Spring Boot and Spring Security along with structured database design. It effectively resolves conflicting requirements by maintaining a balance between strong security mechanisms and user-friendly interfaces. The system also involves analytical design through ER modeling, RESTful API development, and integration of multiple services. Furthermore, it introduces challenges beyond conventional systems, such as intelligent job matching and real-time application tracking. The project considers diverse stakeholder requirements and integrates all system components into a cohesive and scalable solution.

7.2 Mapping of the Project to Complex Engineering Activities (CEA)

EA No.	Attribute	Characteristics	Justification based on the Project
EA1	Range of Resources	Use of diverse technologies	Utilizes Spring Boot, Spring MVC, Spring Data JPA, MySQL/H2, Thymeleaf, Spring Security, and JavaMailSender for full-stack development.
EA2	Level of Interactions	Complex module interactions	Manages interactions between job listings, applications, authentication, resume uploads, and notifications.
EA3	Innovation	Application of modern technologies	Includes role-based dashboards, dynamic filtering, file uploads using MultipartFile, and automated email notifications.
EA4	Consequences to Society	Impact on users	Enhances employment accessibility, reduces manual effort, and ensures a transparent recruitment process.
EA5	Familiarity	Beyond standard systems	Combines authentication, job management, application tracking, and analytics into a unified platform.

Table 7.2: Mapping of Project to Complex Engineering Activities (CEA)

Table 7.2 illustrates the mapping of the system to Complex Engineering Activities (CEA). The project employs a wide range of technologies and tools,

demonstrating effective resource utilization. It involves complex interactions across multiple modules, including authentication, job management, and notification systems. The system showcases innovation through modern features such as dynamic filtering, role-based dashboards, and automated communication. It also contributes positively to society by improving job accessibility and reducing recruitment inefficiencies. Overall, the project extends beyond standard practices by integrating multiple functionalities into a single, scalable platform.

7.3 SDG Mapping (CEP Section)

SDG No.	SDG Title	Relevance to the Project
SDG 8	Decent Work and Economic Growth	Enables employment opportunities by connecting job seekers and employers, supporting structured and fair recruitment.
SDG 9	Industry, Innovation and Infrastructure	Promotes digital transformation through REST APIs, automation, and modern web technologies.
SDG 10	Reduced Inequalities	Provides equal access to job opportunities through a standardized digital platform.
SDG 16	Peace, Justice and Strong Institutions	Ensures secure and transparent processes using authentication and role-based access control.

Table 7.3: SDG Mapping for the Project

Table 7.3 highlights the alignment of the Job Portal Management System with Sustainable Development Goals (SDGs). The system supports SDG 8 by enhancing employment accessibility and economic growth. It contributes to SDG 9 by promoting innovation and digital infrastructure in recruitment. It also addresses SDG 10 by reducing inequalities through an inclusive platform accessible to all users. Additionally, it aligns with SDG 16 by ensuring secure, transparent, and reliable recruitment processes through strong authentication and data protection mechanisms.

References

Platforms and Systems Studied

- 1 LinkedIn: Professional Networking and Job Portal Platform.
<https://www.linkedin.com>
Review Quote: "A widely used platform for professional networking, job search, and recruitment with advanced filtering and recommendation features."
- 2 Indeed: Global Job Search Engine.
<https://www.indeed.com>
Review Quote: "Provides a comprehensive job search experience with extensive listings and easy application processes."
- 3 Naukri.com: Online Recruitment Platform.
<https://www.naukri.com>
Review Quote: "Offers job listings and resume management features widely used in the recruitment industry."

Technical Frameworks and Documentation

- 4 Spring Boot Documentation: Framework for building Java-based applications.
<https://spring.io/projects/spring-boot>
- 5 Spring Security Documentation.
<https://spring.io/projects/spring-security>
- 6 MySQL Documentation: Relational Database Management System.
<https://dev.mysql.com/doc>

7 Thymeleaf Documentation: Server-side Java Template Engine.

<https://www.thymeleaf.org>

8 JavaScript Documentation.

<https://developer.mozilla.org/en-US/docs/Web/JavaScript>

Discussion of References

Various platforms and technologies have contributed to the development of modern job portal systems. LinkedIn provides a professional networking platform with advanced job search and recruitment features [1]. Indeed offers a comprehensive job search engine with extensive listings and simplified application processes [2]. Similarly, Naukri.com provides job listings and resume management functionalities widely used in the recruitment industry [3].

On the technical side, Spring Boot enables the development of scalable and efficient Java-based web applications [4], while Spring Security provides robust authentication and authorization mechanisms [5]. MySQL serves as a reliable relational database for storing system data [6]. Thymeleaf supports server-side rendering for dynamic web pages [7], and JavaScript enhances interactivity and responsiveness of the user interface [8].